Worksheet 5.2 — Trigonometric Ratios & the Unit Circle

Grade 12 Advanced Functions (MHF4U) · Unit 5: Trigonometric Functions

The Big Ideas

On the unit circle the point at angle θ is $(\cos\theta, \sin\theta)$; this gives every special-angle value and the sign in each quadrant.

- $(\cos\theta, \sin\theta)$; $\tan\theta = \frac{\sin\theta}{\cos\theta}$.
- $\sin\frac{\pi}{6} = \frac{1}{2}$, $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$, $\sin\frac{\pi}{4} = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$.
- CAST: which ratios are positive by quadrant.

Core Concepts

The unit circle

On the circle of radius 1, the point at angle θ has coordinates $(\cos\theta,\ \sin\theta)$ the definition that extends trig to every angle.

Exact values

The special angles $\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$ and their multiples give exact coordinates worth knowing all around the circle.

Signs by quadrant

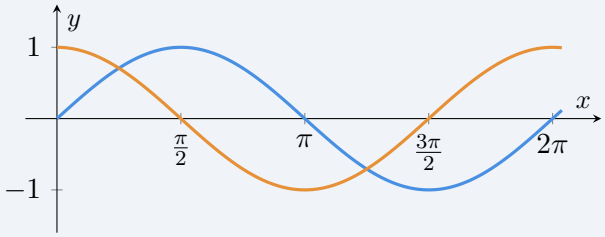
The signs of $\cos\theta$ and $\sin\theta$ follow the point's position, matching the CAST rule.

Worked Examples

Example 1 — Sine

Find $\sin\frac{\pi}{4}$.

Solution. $\frac{\pi}{4} = 45° \Rightarrow \frac{\sqrt{2}}{2}$. Sine and cosine are the same wave shifted by $\frac{\pi}{2}$:



Example 2 — Second quadrant

Find $\cos\frac{3\pi}{4}$.

Solution. Related angle $\frac{\pi}{4}$, and $\cos < 0$ in QII: $-\frac{\sqrt{2}}{2}$.

Example 3 — Quadrantal

Find $\sin\frac{3\pi}{2}$.

Solution. Point $(0, -1) \Rightarrow -1$.

Example 4 — Cosine of 2

Find $\cos 2\pi$.

Solution. Point $(1, 0) \Rightarrow 1$.

Example 5 — Tangent

Find $\tan \frac{\pi}{3}$.

Solution. $\frac{\sqrt{3}/2}{1/2} = \sqrt{3}$.

Example 6 — Sine

Find $\sin \frac{2\pi}{3}$.

Solution. Related angle $\frac{\pi}{3}$, $\sin > 0$ in QII: $\frac{\sqrt{3}}{2}$.

Example 7 — Cosine

Find $\cos \frac{5\pi}{3}$.

Solution. Related angle $\frac{\pi}{3}$, $\cos > 0$ in QIV: $\frac{1}{2}$.

Example 8 — Sine

Find $\sin \frac{7\pi}{6}$.

Solution. Related angle $\frac{\pi}{6}$, $\sin < 0$ in QIII: $-\frac{1}{2}$.

Example 9 — Tangent

Find $\tan \frac{3\pi}{4}$.

Solution. $\frac{\sin}{\cos} = \frac{\sqrt{2}/2}{-\sqrt{2}/2} = -1$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $\cos\frac{\pi}{4}$.

Question 2

Find $\sin\frac{5\pi}{3}$.

Question 3

Find $\cos\frac{3\pi}{2}$.

Question 4

Find $\sin 0$.

Question 5

In which quadrant is $\sin\theta < 0,\ \cos\theta > 0$?

Question 6

Find $\tan \frac{\pi}{6}$.

Question 7

Find $\cos \frac{7\pi}{6}$.

Question 8

Find $\sin \frac{5\pi}{4}$.

Question 9

Find $\tan \frac{5\pi}{6}$.

Question 10

Find $\cos 0$.

Question 11

Find $\tan \pi$.

Question 12

In which quadrant are both $\sin\theta < 0$ and $\cos\theta < 0$?

Question 13 — Challenge

Using the unit circle, find $\sin \frac{11\pi}{6}$ and $\cos \frac{11\pi}{6}$.

Question Recap & Answer Key

Question Recap

1. Find $\cos\frac{\pi}{4}$.
2. Find $\sin\frac{5\pi}{3}$.
3. Find $\cos\frac{3\pi}{2}$.
4. Find $\sin 0$.
5. In which quadrant is $\sin\theta < 0,\ \cos\theta > 0$?
6. Find $\tan\frac{\pi}{6}$.
7. Find $\cos\frac{7\pi}{6}$.
8. Find $\sin\frac{5\pi}{4}$.
9. Find $\tan\frac{5\pi}{6}$.
10. Find $\cos 0$.
11. Find $\tan\pi$.
12. In which quadrant are both $\sin\theta < 0$ and $\cos\theta < 0$?
13. Using the unit circle, find $\sin\frac{11\pi}{6}$ and $\cos\frac{11\pi}{6}$.

✓ Answer Key

1. $\frac{\sqrt{2}}{2}$
2. $-\frac{\sqrt{3}}{2}$
3. 0
4. 0
5. IV
6. $\frac{1}{\sqrt{3}}$
7. $-\frac{\sqrt{3}}{2}$
8. $-\frac{\sqrt{2}}{2}$
9. $-\frac{1}{\sqrt{3}}$
10. 1
11. 0
12. III
13. $\sin = -\frac{1}{2},\ \cos = \frac{\sqrt{3}}{2}$